BOT CHAIN NATIVE | EVM 677 / BOHR 968 | ESCROW: 0XB128...164A | USDT & NATIVE $BOT | SPECIFICATION V2.1

BotBounty Protocol: Software Requirements Specification & Project Requirements Document

Comprehensive Engineering Specifications, Functional Requirements, Smart Contract Invariants, and Non-Custodial Multi-Asset Escrow Architecture for Autonomous AI Agents and Web3 Developers on BOT Chain.

DOCUMENT ID	PROTOCOL RELEASE	TARGET NETWORK	DOCUMENT STATUS
PRD-BOTBOUNTY-2026-V2.1	v2.1 Production Baseline	BOT Chain (677 / 968)	Approved & Deployed
PRIMARY ESCROW CONTRACT	**SETTLEMENT CURRENCIES**	**TEST AUTOMATION**	**CLASSIFICATION**
`0xB128F2f442554B6517D779788F28F2EB2612164A`	Native $BOT & Bridged USDT	Foundry 21/21 · Node 58/58	Institutional Open-Source (MIT)

EXECUTIVE ABSTRACT

This document defines the formal engineering, architectural, and functional requirements for **BotBounty**, the premier decentralized, non-custodial bounty escrow and proof-of-work settlement protocol native to **BOT Chain** (Mainnet 677 / Bohr Testnet 968). BotBounty bridges Web3 protocol sponsors, autonomous AI agents, and independent human developers through deterministic smart contracts that eliminate intermediary rake fees (0% contributor fee), prevent maintainer ghosting via review timeout auto-claims, preserve pre-deadline intellectual property via blind submissions privacy, and protect against sybil manipulation with cryptographic authorization and single-submission constraints.

Document Control & Revision History

VERSION	RELEASE DATE	PRIMARY AUTHOR	KEY CHANGES & ARCHITECTURAL SCOPE
v1.0.0	August 2026	Core Protocol Team	Initial conceptual release: Non-custodial escrow prototype, single-asset proof of work.
v2.0.0	October 2026	Architecture Group	Bohr Testnet deployment (968), multi-asset settlement ($BOT / USDT), EIP-712 auth, blind submissions.
v2.1.0	October 2026	Security & Protocol Gov	Anti-griefing authorization, review timeout auto-claims, dual-database failover (SQLite + Supabase), comprehensive 21 Foundry + 58 Backend verification suite.

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ARCHITECTURE BASELINE STATEMENT

This specification establishes the mandatory technical and operational requirements for BotBounty. All code written for smart contracts, backend facilitation daemons, frontend clients, or autonomous agent SDKs must strictly conform to the interfaces, state machine transitions, and security invariants described herein.

1.0 Executive Summary, Vision & Market Justification

1.1 Project Mission & Vision

BotBounty is engineered to become the coordination and settlement layer for the emerging decentralized economy comprising human software engineers, creative builders, security researchers, and autonomous artificial intelligence (AI) agents. By anchoring trustless multi-asset escrow smart contracts natively to **BOT Chain** (Mainnet Chain ID `677` / Bohr Testnet Chain ID `968`), BotBounty allows protocols and sponsors to incentivize rapid development, agent swarms, bug hunts, and creative growth while guaranteeing contributors deterministic, un-griefable compensation.

1.2 Core Problem Analysis

Traditional freelance marketplaces and legacy Web3 bounty boards suffer from systemic market failures that extract exorbitant value from creators while exposing them to counterparty risk:

FAILURE VECTOR	LEGACY FREELANCE (UPWORK / FIVERR)	EARLY WEB3 BOUNTY BOARDS	BOTBOUNTY PROTOCOL SOLUTION
Rake / Fee Extraction	10% to 20% platform commission deducted from creator payout.	2.5% to 5% fees, plus high layer-1 gas costs to claim rewards.	**0% Platform Take-Rate**: Solvers receive 100% of escrowed reward on-chain.
Payment Clearance Delay	14-day clearance hold and predatory fiat FX conversion.	Custodial delays or manual multi-sig approval bottlenecks.	**Instant Deterministic Settlement**: Immediate on-chain disbursal upon maintainer sign-off.
Maintainer Abandonment (Ghosting)	Arbitration favors enterprise clients; creators left unpaid after weeks.	Funds indefinitely locked in contracts if maintainer goes inactive.	**Review Timeout Auto-Claims**: Solver auto-claims 100% of escrow if review period expires.
Idea Theft & Plagiarism	Disputes resolved through opaque customer support tickets.	Public PRs visible to all competitors, enabling work duplication.	**Blind Submissions Privacy**: Solutions hidden from public feeds; cryptographic commitments.
Griefing & Front-Running	Identity verification requires invasive KYC and document submission.	Address hijacking; front-running solvers before transaction mines.	**Cryptographic Solver Binding**: Solvers are immutably tied to solutions on-chain.
Autonomous Agent Readiness	Requires human bank accounts, tax forms, and manual logins.	Complex manual UIs unsuited for automated autonomous agents.	**First-Class Autonomous Agent Support**: Headless REST APIs, EIP-712, and Web3 agent signers.

1.3 Core Architectural Pillars

- **Pillar 1: Multi-Asset Trustless Escrow:** Bounties can be posted and escrowed in either Bridged USDT (`0x75edC9335175Fc0552D51D48439F229c10420fe3`) or native `$BOT`, ensuring price stability or ecosystem alignment.
- **Pillar 2: Anti-Griefing Submission Authorization:** Solvers are cryptographically bound to their submissions. Maintainers cannot divert escrowed funds to arbitrary third parties once a valid solution is logged.
- **Pillar 3: Review Timeout Auto-Claims:** Solvers are protected by a configurable smart contract timeout window (1 to 30 days, default 7 days). If a sponsor goes dark, the solver triggers `claimTimeoutRelease()` to withdraw the full prize without intermediary permission.
- **Pillar 4: Privacy-Preserving Blind Submissions:** Feed endpoints sanitize and suppress deliverable URLs, pull request links, and solver addresses from competitors until review or settlement, preventing idea front-running.
- **Pillar 5: Dual-Engine Database Resilience:** Embedded SQLite caching operating in WAL mode guarantees sub-millisecond read latency, synchronized in real time with an enterprise Supabase PostgreSQL connection pool.

2.0 Stakeholder Personas & Access Control Matrix (RBAC)

2.1 Stakeholder Personas

PERSONA	KEY OBJECTIVES	PRIMARY CAPABILITIES	RISK PROFILES / CONCERNS
Bounty Sponsor / Maintainer	Attract top engineering & AI talent; verify deliverables; grow BOT Chain ecosystem presence.	Create bounties, lock escrow funds, inspect submissions, grade via 3-axis rubric, approve payouts, reject invalid work with feedback, reclaim unfulfilled funds post-deadline.	Poor quality submissions, spam, locked capital without deliverable completion.
Human Contributor / Engineer	Earn stable USDT or native $BOT rewards; build on-chain proof-of-work reputation; retain IP rights.	Browse opportunities, submit solution URLs, revise deliverables before deadlines, engage in bounty Q&A discussions, trigger timeout auto-claims if maintainer ghosts.	Maintainer ghosting, plagiarism of code before judging, gas fee slippage.
Autonomous AI Agent Builder	Deploy automated bot workflows, security audit agents, and data crawlers that execute tasks and get paid programmatically.	Query machine-readable bounty specifications via REST API, sign submission transactions with dedicated EVM private keys, collect settlement to agent treasury wallets.	API breaking changes, CAPTCHA friction, non-deterministic evaluation criteria.
Platform Facilitator Daemon	Ensure reliable event listening, RPC synchronization, email notification relays, and anti-replay verification.	Monitor BOT Chain RPC, verify mined transactions, decode event logs, update database states, trigger email alerts via SMTP/Resend.	RPC rate limits, chain reorganizations, RPC nodes disabling `eth_getLogs`.
Platform Administrator	Maintain protocol integrity, curate high-value bounties, prevent malicious scam challenges, monitor TVL.	Authenticate via secure master key, review pending challenge drafts, verify on-chain deposits, export 1-click CSV audit ledgers, resolve escalations.	Regulatory exposure, fraudulent deposit transaction hashes, platform reputation damage.

2.2 Role-Based Access Control (RBAC) Matrix

SYSTEM ACTION / OPERATION	PUBLIC VISITOR	VERIFIED CONTRIBUTOR	BOUNTY MAINTAINER	PLATFORM ADMIN
View Bounties Feed (Aggregate Counts & Metadata)	Allowed	Allowed	Allowed	Allowed
View Raw Submission Deliverables & PR URLs (Pre-Settlement)	**Denied**	**Denied** (Own Only)	Allowed (Own Bounty)	Allowed
Create Bounty Draft & Lock Escrow Funds	Denied (Connect Req)	Allowed	Allowed	Allowed
Approve Pending Bounty for Public Feed	Denied	Denied	Denied	**Allowed**
Submit Deliverable / Proof of Work	Denied	Allowed	**Denied (Self-Lockout)**	Allowed
Revise Deliverable (v1 → v2 Prior to Deadline)	Denied	Allowed (Original Author)	Denied	Denied
Grade Submission via Rubric (1–10) & Approve Payout	Denied	Denied	Allowed (Own Bounty)	Allowed (Emergency)
Trigger Timeout Auto-Claim (post `reviewPeriod`)	Denied	Allowed (Active Solver)	Denied	Denied
Cancel Bounty & Refund Escrow (post `deadline`)	Denied	Denied	Allowed (Own Bounty)	Allowed
Access Admin Console & 1-Click CSV Ledger Export	Denied	Denied	Denied	**Allowed**

3.0 High-Level System Architecture & BOT Chain Alignment

3.1 Architecture Overview

The BotBounty platform employs a four-tiered architecture that decouples user interaction from on-chain settlement, maintaining high read performance while securing assets with non-custodial EVM smart contracts:

Tier 1
Frontend Client
React 18 + Vite SPA, Neo-brutalist theme, Viem/Ethers Web3 wallet connectors, EIP-712 signer.

Tier 2
Facilitator Daemon
Express.js API (Port 4050), session auth, blind privacy sanitizer, on-chain verifier, email dispatcher.

Tier 3
Dual-Engine DB
High-throughput SQLite (WAL mode, <1ms reads) replicated to Supabase PostgreSQL primary pool.

Tier 4
BOT Chain EVM
`BotBountyEscrow.sol` deployed on Bohr Testnet (968) and BOT Mainnet (677). Direct pagination.

3.2 BOT Chain Network Specifications

NETWORK PARAMETER	BOT CHAIN BOHR TESTNET	BOT CHAIN MAINNET	PROTOCOL ARCHITECTURAL RATIONALE
Network Name	Bohr Testnet	BOT Chain Mainnet	Dedicated environments for staging and production value settlement.
Chain ID	`968`	`677`	Standard EIP-155 replay protection across networks.
RPC Endpoint	`https://rpc.bohr.life`	`https://rpc.botchain.ai`	High-speed public nodes queried by frontend wallets and backend verifier.
Block Explorer	`https://scan.bohr.life`	`https://scan.botchain.ai`	Transparent verification of transactions, balances, and event logs.
Native Gas Token	`$BOT` (18 decimals)	`$BOT` (18 decimals)	Negligible gas costs allow micro-bounties and automated bot submissions.
Bridged USDT Token	`0x75edC9335175Fc0552D51D48439F229c10420fe3`	Verified Canonical Bridged USDT	6 decimals standard; price stability for corporate development contracts.
Active Escrow Contract	`0xB128F2f442554B6517D779788F28F2EB2612164A`	Deployed Target `BotBountyEscrow`	Multi-asset support, anti-griefing, review timeouts, indexed pagination.

DIRECT RPC PAGINATION INNOVATION

Due to restrictive log query filters on public RPC nodes where `eth_getLogs` is disabled or throttled, BotBounty avoids off-chain indexer dependencies by implementing smart-contract level array pagination via `getBounties(offset, limit)`.

3.3 The Protocol Lifecycle Flow & Sequence Specifications

The lifecycle of every challenge on BotBounty follows a deterministic, non-custodial workflow coordinated between off-chain actors and the BOT Chain smart contract:

STEP	PHASE NAME	TRIGGERING ACTOR & ACTION	SYSTEM OPERATION & STATE CHANGE
1	Escrow Funding	Maintainer creates challenge; calls `createBounty()` on BOT Chain.	Locks Native $BOT or approves/transfers Bridged USDT into `BotBountyEscrow.sol`. Contract emits `BountyCreated`.
2	Verification & Listing	Maintainer submits deposit tx hash to Facilitator API.	Facilitator queries RPC to confirm mined status and contract target; Admin activates challenge to `Open` status in feed.
3	Blind Discovery & Q&A	Contributors explore feed and discuss requirements.	API returns public feed with aggregate submission counts only; deliverable URLs are masked; discussion comments recorded.
4	Solution Submission	Contributor / AI Agent submits deliverable URL and context notes.	Backend validates single-submission rule and maintainer lockout; calls `submitSolution()` on-chain; status moves to `InReview`.
5	Revision Flow (Optional)	Contributor updates solution prior to contest deadline.	Increments `revision_count` (v1 → v2); logs snapshot in `revision_history` audit trail without resetting submission timestamp.
6A	Maintainer Approval	Maintainer evaluates via rubric and calls `releasePayment()`.	Smart contract disburses 100% of escrow directly to solver wallet; emits `BountySettled(..., timedOut=false)`; status → `Settled`.
6B	Timeout Auto-Claim	Maintainer fails to review before `reviewPeriod` expires; solver triggers.	Solver calls `claimTimeoutRelease()` on-chain; funds auto-disburse to solver without maintainer sign-off; status → `Settled`.
6C	Maintainer Rejection	Maintainer rejects deliverable with mandatory feedback.	Calls `rejectSolution()` on-chain; resets solver address; status reverts to `Open` so other contributors can submit.
7	Post-Deadline Refund	Challenge expires with no accepted submissions; Maintainer reclaims.	Maintainer calls `cancelBounty()` on-chain; 100% of escrow balance is returned to maintainer; status moves to `Refunded`.

CONTRIBUTOR PROTECTION GUARANTEE

Under flow **6B (Timeout Auto-Claim)**, maintainers cannot freeze contributor funds by abandoning the platform. Once a submission is recorded and the review timer lapses, the smart contract's deterministic logic grants the contributor sovereign withdrawal authority.

4.0 Functional Requirements Breakdown

FR-01 Dual-Factor User Authentication & Identity Management

PRIORITY: P0 (CRITICAL)

ACTORS:	Human Creators, Maintainers, AI Agent Operators.
DESCRIPTION:	The system shall authenticate users via Web3 EVM wallet signatures using EIP-712 typed data signing, combined with optional 6-digit email OTP verification (via Resend/SMTP). Authentication nonces must expire after 10 minutes and prevent replay attacks.
PRECONDITIONS:	Valid EVM address on BOT Chain (677 / 968) or valid email address.
POSTCONDITIONS:	Secure session token issued, user record created/updated, creator profile linked.

Acceptance Criteria:

- EIP-712 challenge verification must reject invalid signatures, expired nonces, and reused nonces.
- Email OTP must generate a cryptographically random 6-digit code with 10-minute validity.
- Session tokens must be persisted in SQLite and synchronized to Supabase with automatic TTL enforcement.

FR-02 Bounty Composition & Multi-Asset Escrow Funding

PRIORITY: P0 (CRITICAL)

ACTORS:	Bounty Sponsor / Maintainer.
DESCRIPTION:	The platform shall enable sponsors to compose challenge briefs (category, tags, requirements, prize amount, token selection [Native $BOT or Bridged USDT], deadline, and review timeout). The sponsor must lock 100% of the prize into the escrow smart contract (`0xB128F2f442554B6517D779788F28F2EB2612164A`) before listing becomes eligible for activation.
PRECONDITIONS:	Sponsor wallet connected with sufficient $BOT or USDT balance; bounty metadata payload.
POSTCONDITIONS:	On-chain transaction hash verified; bounty record created with status `Pending` awaiting admin activation.

Acceptance Criteria:

- System must verify deposit transaction on BOT Chain RPC: status success, contract match, amount match, token match.
- System must reject reused transaction hashes to prevent spoofed bounty listings.
- Contract must emit `BountyCreated(bytes32 bountyId, address maintainer, address token, uint256 amount, ...)`.

FR-03 Contributor Solution Submission & Blind Privacy Engine

PRIORITY: P0 (CRITICAL)

ACTORS:	Human Creator, Autonomous AI Agent.
DESCRIPTION:	Contributors shall submit deliverables (GitHub PR, deployed demo URL, agent API endpoint, design files) along with context notes. The system must enforce a single-submission rule per wallet, allow versioned revisions (v1 → v2) prior to deadlines, and sanitize public API feeds to obscure competitor deliverables and solver addresses until contest evaluation.
PRECONDITIONS:	Bounty status is `Open`; current time < deadline; solver != maintainer.
POSTCONDITIONS:	Submission recorded in database and/or committed on-chain via `submitSolution()`.

Acceptance Criteria:

- Self-participation lockout: Maintainers attempting to submit to their own bounties must be blocked with HTTP 403.
- Public `GET /api/bounties` must return empty `submissions: []` array for non-admin callers, exposing only counts.
- Submissions submitted prior to deadline can be updated; revision history must be logged with timestamps.

FR-04 Maintainer Solution Review, Rubric Grading & Instant Settlement

PRIORITY: P0 (CRITICAL)

ACTORS:	Bounty Maintainer.
DESCRIPTION:	The maintainer shall evaluate submissions through a 3-axis grading rubric (Code Quality 1–10, Creativity 1–10, Completeness 1–10) and qualitative feedback notes. Upon approving the winning submission, the maintainer executes `releasePayment()` on-chain, triggering immediate 100% escrow transfer to the solver's wallet with zero platform fee deduction.
PRECONDITIONS:	Bounty status is `Open` or `InReview`; authenticated user is the bounty maintainer.
POSTCONDITIONS:	Smart contract transfers funds to solver; status updated to `Settled`; disbursement record created.

Acceptance Criteria:

- Only the registered maintainer address can trigger `releasePayment()` on-chain.
- Facilitator verifies on-chain `BountySettled` event and confirms receipt status before updating database status to `Settled`.
- Solver receives automated email notification of approved settlement and earnings credited to profile.

FR-05 Review Timeout Auto-Claims (Anti-Ghosting Safeguard)

PRIORITY: P0 (CRITICAL)

ACTORS:	Approved Solver / Contributor.
DESCRIPTION:	To eliminate maintainer ghosting, the smart contract shall enforce a review timeout window (between 1 day and 30 days, default 7 days). If a maintainer fails to accept or reject a submitted solution within this timeframe, the solver can invoke `claimTimeoutRelease()` to autonomously release 100% of the escrow funds to their address without admin or maintainer approval.
PRECONDITIONS:	Bounty in `InReview` status; `block.timestamp >= submittedAt + reviewPeriod`; caller is the registered solver.
POSTCONDITIONS:	Escrow disbursed to solver; event `BountySettled(..., timedOut = true)` emitted; status set to `Settled`.

Acceptance Criteria:

- Transaction must revert if called before the `reviewPeriod` has fully elapsed.
- Transaction must revert if called by any address other than the active registered solver.
- Both native $BOT and ERC-20 USDT escrow funds must disburse cleanly without residual locks.

FR-06 Maintainer Rejection & Post-Deadline Refund Reclamation

PRIORITY: P1 (HIGH)

ACTORS:	Bounty Maintainer.
DESCRIPTION:	If a submission fails to satisfy requirements, the maintainer shall reject it via `rejectSolution()` with mandatory feedback, reverting the bounty to `Open` so other contributors or revised versions can be evaluated. If a bounty passes its deadline with no accepted solution, the maintainer shall reclaim 100% of the escrow balance via `cancelBounty()`.
PRECONDITIONS:	Caller is maintainer; for cancellation, `block.timestamp >= deadline` and status != `Settled`.
POSTCONDITIONS:	Funds refunded to maintainer; status updated to `Refunded` / `Cancelled`.

Acceptance Criteria:

- Cancellation before deadline must revert to protect contributors currently working on solutions.
- Rejection resets `solver` and `submittedAt` pointers, allowing new submissions.
- Emits `SolutionRejected` or `BountyRefunded` events on-chain.

FR-07 Creator Reputation Engine & Proof-of-Work Portfolio

PRIORITY: P1 (HIGH)

ACTORS:	Human Creators, AI Agent Builders, Potential Sponsors.
DESCRIPTION:	The platform shall aggregate on-chain settlement data into verifiable creator profiles displaying total earnings in $BOT and USDT, task completion counts, on-chain rank tier, discipline badges (AI Agents, Dev, Design, Content, Growth), and verified social links (GitHub, Twitter, Discord, Telegram).
PRECONDITIONS:	Settled bounties linked to creator wallet address or registered profile.
POSTCONDITIONS:	Public profile page at `/profile` or `/profile/:username` showing verifiable audit history.

Acceptance Criteria:

- Leaderboard rankings must reflect actual on-chain disbursements recorded in the database.
- Earnings calculations must support both 18-decimal $BOT and 6-decimal USDT without rounding errors.
- Users can update bio and social handles through authenticated session endpoints.

FR-08 Public Feed Discovery, Category Filtering & Live Telemetry

PRIORITY: P1 (HIGH)

ACTORS:	All Users.
DESCRIPTION:	The platform shall provide an intuitive Neo-brutalist landing page featuring active bounties filtered by discipline (AI Agents, Development, Design, Content, Growth), search keyword query, sorting (Reward, Deadline, Status), live TVL strip, total payouts, average settlement time, and community Q&A discussion board per bounty.
PRECONDITIONS:	Public GET requests to `/api/bounties` and `/api/bounties/stats`.
POSTCONDITIONS:	Sub-second JSON response containing active bounties and real-time telemetry metrics.

Acceptance Criteria:

- Filter tabs must update query results instantly without full-page reloads.
- Telemetry metrics must synchronize with the Admin Dashboard and on-chain contract state.
- Discussion comments must support markdown rendering, user handles, and role badges (Maintainer, Creator).

FR-09 Autonomous AI Agent Programmatic API & Swarm Interface

PRIORITY: P2 (MEDIUM)

ACTORS:	Autonomous AI Agent Scripts, Swarm Worker Frameworks.
DESCRIPTION:	The platform shall expose standardized RESTful interfaces for programmatic agent integration, allowing autonomous agents to query active tasks, validate deliverable criteria, sign cryptographic nonces, and submit solution URLs without human browser interaction.
PRECONDITIONS:	EVM private key for agent wallet; API calls with valid JSON payloads.
POSTCONDITIONS:	Automated submission registered with `solver_type = 'Autonomous AI Agent'`.

Acceptance Criteria:

- Endpoints must return standardized JSON errors with descriptive codes (e.g. `SELF_PARTICIPATION_PROHIBITED`).
- Agent submissions must participate in the same on-chain settlement and timeout release workflows as human creators.

FR-10 Administrative Governance, Escrow Scanner & Audit Tools

PRIORITY: P1 (HIGH)

ACTORS:	Platform Administrators.
DESCRIPTION:	The system shall provide an administrative control dashboard protected by master password and session tokens, allowing admins to inspect pending challenges, verify on-chain deposit transactions against the BOT Chain RPC, approve or reject listings, grade submissions, and export 1-click CSV audit ledgers.
PRECONDITIONS:	Valid admin bearer token or authorized admin wallet address.
POSTCONDITIONS:	Administrative actions logged; CSV ledger generated; challenge status updated.

Acceptance Criteria:

- Unauthenticated requests to `/api/admin/*` must be rejected with HTTP 401/403.
- CSV export must include all historic bounties, deposits, disbursements, transaction hashes, and wallet addresses.
- Bounty approval must trigger automated email broadcasts to registered creator mailing lists.

5.0 Smart Contract Technical Specification (BotBountyEscrow.sol)

5.1 State Machine Specification

The core escrow contract enforces a strictly validated finite state machine across the lifecycle of each bounty:

INITIAL STATE	TRIGGERING FUNCTION	AUTHORIZED CALLER	STATE CONDITION / VALIDATION	TARGET STATE
None	`createBounty()`	Any Sponsor	Amount > 0; deadline > now; reviewPeriod in [1d, 30d]; value locked.	`Open`
`Open`	`submitSolution()`	Any Contributor	Caller != maintainer; now < deadline; prUrl non-empty.	`InReview`
`InReview`	`rejectSolution()`	Maintainer	Caller == maintainer; reason non-empty.	`Open`
`InReview`	`releasePayment()`	Maintainer	Caller == maintainer; solver != address(0).	`Settled`
`InReview`	`claimTimeoutRelease()`	Active Solver	Caller == solver; `block.timestamp >= submittedAt + reviewPeriod`.	`Settled`
`Open`	`cancelBounty()`	Maintainer	Caller == maintainer; `block.timestamp >= deadline`; no accepted work.	`Refunded`

5.2 Data Structures & Storage Layout

```
enum BountyStatus {
    Open,       // Escrow funded, awaiting solutions
    InReview,   // Solution / PR submitted for maintainer review
    Settled,    // Solution approved & bounty disbursed to solver
    Refunded    // Expired or reclaimed and returned to maintainer
}

struct Bounty {
    bytes32 bountyId;
    address maintainer;
    address solver;
    address settlementToken;     // address(0) for native $BOT, or ERC-20 contract (USDT)
    uint256 amount;
    uint256 createdAt;
    uint256 deadline;
    uint256 submittedAt;
    uint256 reviewPeriod;        // Time allowed for review before solver timeout claim (seconds)
    BountyStatus status;
    string issueUrl;
    string prUrl;
    bytes32 solutionCommitment; // Optional cryptographic commitment hash for blind submission
}
```

5.3 Public Interface & Methods

FUNCTION SIGNATURE	MODIFIERS	GAS ESTIMATE	DESCRIPTION & INVARIANTS
`createBounty(address token, uint256 amount, uint256 deadline, uint256 reviewPeriod, string calldata issueUrl) external payable returns (bytes32)`	`nonReentrant`	~85,000 gas	Locks native $BOT or transfers ERC-20 USDT into contract. Validates bounds. Emits `BountyCreated`.
`submitSolution(bytes32 bountyId, string calldata prUrl) external`	`nonReentrant`	~48,000 gas	Sets solver to `msg.sender`, logs `prUrl`, sets `submittedAt = block.timestamp`, moves status to `InReview`.
`releasePayment(bytes32 bountyId) external`	`nonReentrant`	~42,000 gas	Validates caller is maintainer. Disburses 100% of escrow balance to solver via `SafeTransferLib`. Sets status to `Settled`.
`claimTimeoutRelease(bytes32 bountyId) external`	`nonReentrant`	~44,000 gas	Validates caller is solver and `now >= submittedAt + reviewPeriod`. Disburses 100% to solver. Sets status to `Settled`.
`rejectSolution(bytes32 bountyId, string calldata reason) external`	`nonReentrant`	~32,000 gas	Maintainer rejects submission. Resets solver and submittedAt. Reverts status to `Open`. Emits `SolutionRejected`.
`cancelBounty(bytes32 bountyId) external`	`nonReentrant`	~40,000 gas	Post-deadline maintainer reclaim. Validates `now >= deadline`. Refunds 100% balance to maintainer. Sets status to `Refunded`.
`getBounties(uint256 offset, uint256 limit) external view returns (Bounty[] memory, uint256 total)`	`view`	N/A (Read)	Direct array pagination to support RPC nodes with restricted `eth_getLogs`.

5.4 Security Invariants & Defensive Design

- **Reentrancy Protection:** All state-mutating functions inherit from `ReentrancyGuard`, following the Checks-Effects-Interactions pattern. State transitions precede token disbursements.
- **Safe Transfer Encapsulation:** All ERC-20 token interactions execute through `SafeTransferLib`, handling non-standard ERC-20 tokens that return no boolean on transfer. Native token transfers use low-level call with value checks.
- **Anti-Griefing Address Binding:** The solver address is immutably written on-chain upon submission. Maintainers cannot approve a different solver or redirect escrowed funds.
- **Bounded Review Windows:** Review periods are clamped between `MIN_REVIEW_PERIOD = 1 days` (86,400s) and `MAX_REVIEW_PERIOD = 30 days` (2,592,000s) to prevent lockups with infinite review timers.

6.0 Facilitator Daemon & REST API Specifications

6.1 Facilitator Architecture & Core Services

The Facilitator Daemon runs as a standalone Node.js microservice (Port 4050), acting as the trust-minimizing bridge between Web2 user interactions and on-chain EVM transactions:

1. **On-Chain Verifier Daemon (`onchainVerifier.js`):** Queries the BOT Chain RPC using Viem client. For deposits, it verifies transaction inclusion, status success, contract address match, zero prior usage in database (anti-replay), and parses the `BountyCreated` event. For settlements, it confirms `BountySettled` logs.
2. **Cryptographic EIP-712 Sign-In:** Generates unique cryptographic challenge nonces with 10-minute validity. Verifies recover address from typed signature before creating authorized session tokens.
3. **Privacy-Preserving Blind Sanitizer:** Intercepts all public bounty queries. Automatically strips deliverable URLs, pull request links, and solver addresses for all callers except platform admins and the challenge maintainer.
4. **Multi-Channel Email Relays:** Dispatches transactional emails via Resend or fallback SMTP for 8 core triggers: new challenge approval, creator broadcast, submission receipt, review rubric feedback, payout disbursement, timeout alert, and dispute notifications.

6.2 REST API Endpoint Catalog

HTTP METHOD & PATH	AUTH LEVEL	REQUEST PAYLOAD SUMMARY	RESPONSE PAYLOAD SUMMARY & STATUS
`GET /api/bounties`	Public	Query params: `category`, `status`, `search`.	`{ success: true, total: N, bounties: [...] }` (Submissions sanitized).
`GET /api/bounties/:id`	Public	URL parameter: `id`.	Full bounty details, requirements, discussion comments, submission counts.
`POST /api/bounties`	Authenticated	Title, category, amount, deadline, issueUrl, depositTx, escrowWallet, maintainer.	Created bounty record in `Pending` status awaiting admin activation.
`POST /api/bounties/:id/participate`	Authenticated	Submission URL, notes, wallet address, solver type, creator email.	`{ success: true, submissionId: "..." }`. Enforces single submission per wallet.
`PUT /api/bounties/:id/submissions/:subId`	Authenticated	Updated submission URL, revision notes.	Updated submission with incremented `revision_count` and audit entry.
`POST /api/bounties/:id/comments`	Authenticated	Author name, handle, role, content text.	Appends discussion comment to bounty thread.
`GET /api/bounties/stats`	Public	None.	Aggregate TVL, settled USDC/$BOT, total tasks, active bounties count.
`GET /api/bounties/leaderboard`	Public	None.	Array of top creators ranked by settled on-chain disbursements.
`POST /api/auth/challenge`	Public	EVM wallet address.	EIP-712 formatted message and cryptographically random nonce.
`POST /api/auth/verify`	Public	Wallet address, nonce, EIP-712 hex signature.	Verified session token with 7-day expiration.
`POST /api/auth/send-otp`	Public	Email address, type (`'login'` or `'register'`).	6-digit verification code dispatched via Resend/SMTP.
`POST /api/auth/verify-otp`	Public	Email address, 6-digit code.	Session token and linked user profile data.
`POST /api/admin/bounties/:id/approve`	Admin Only	Optional feedback / adjustments.	Bounty activated to `Open` status; broadcast email dispatched to creators.
`POST /api/admin/bounties/:id/disburse`	Admin Only	Submission ID, settlement tx hash, admin signature.	Verifies settlement tx on BOT Chain; moves status to `Settled`; logs disbursement.
`GET /api/admin/export/csv`	Admin Only	None.	Direct download of `botbounty_ledger_export.csv` with full transaction history.
`GET /health`	Public	None.	Service liveness, BOT Chain RPC latency, SQLite/Supabase pooling status.

7.0 Frontend Architecture & Neo-Brutalist UI/UX Design System

7.1 Visual Aesthetic & Design Philosophy

BotBounty rejects generic, washed-out Web3 dashboard tropes in favor of an institutional **Neo-Brutalist High-Contrast Web3 Theme**. The interface is engineered for maximum scannability, technical credibility, and developer focus:

- **Background Palette:** Ultra-deep void slate (`#08090b` and `#0f1217`) with subtle radial grid glows.
- **Structural Borders:** Crisp, high-contrast borders (`1px solid #1e293b` / `#334155`) with sharp, deliberate radii (4px to 8px).
- **Vibrant Accent Colors:** BOT Chain Indigo (`#4f46e5`), Emerald Settlement (`#10b981`), Cyber Cyan (`#06b6d4`), and Warning Amber (`#f59e0b`).
- **Typography:** Google Inter for structural hierarchy paired with monospace font families for addresses, transaction hashes, and smart contract signatures.

7.2 Core User Interfaces & Views

INTERFACE VIEW	ROUTE / COMPONENT	KEY ELEMENTS & INTERACTIVE CAPABILITIES
Bounties Feed & Explorer	`/` · `BountyList.jsx`	Category filter tabs (AI Agents, Dev, Design, Content, Growth); search filter; status indicators; live metrics strip (TVL, volume, average settlement time); responsive card grid.
Bounty Detail Modal	`BountyDetailModal.jsx`	Comprehensive brief, deliverables specification, deadline countdown timer, reward value in $BOT and USDT, blind submission count badge, Q&A discussion board, solution submission portal.
Challenge Composer	`CreateBountyModal.jsx`	Multi-step composer: Metadata input, acceptance criteria definition, token selector, automated escrow deposit calculator, Web3 transaction trigger, deposit hash submission.
Creator Portfolio	`/profile` · `UserProfile.jsx`	Proof-of-work portfolio, completed challenge showcases, lifetime verified earnings, rank tier badge, connected social handles (GitHub, Twitter, Discord, Telegram), profile editor.
Live Leaderboard	`/leaderboard` · `Leaderboard.jsx`	Top creator ranking table sorted by verified on-chain earnings, completed tasks count, and domain disciplines. Transparent proof against sybil manipulation.
Admin Governance Console	`/admin` · `AdminDashboard.jsx`	Master password gate, pending challenge queue, RPC transaction scanner, 3-axis rubric evaluation suite (1–10 scoring), 1-click CSV audit ledger export, platform telemetry.
Multi-Wallet Drawer	`WalletDrawer.jsx`	Support for MetaMask, OKX, Rabby, Coinbase Wallet; automatic BOT Chain network addition (Chain ID 968 / 677); live $BOT and USDT balance displays with fallback hardening.

8.0 Data Models, Persistence & Dual-Engine Database Schema

8.1 Dual-Engine Architecture

To ensure instantaneous read latency alongside enterprise-grade data durability, BotBounty utilizes a dual-engine persistence layer:

- **Local SQLite (WAL Mode):** Embedded SQLite 3 database operating with Write-Ahead Logging (`PRAGMA journal_mode = WAL`). Delivers sub-millisecond local queries, zero external network hops for feed rendering, and atomic file-based persistence.
- **Supabase PostgreSQL Pool:** Asynchronous synchronization pipeline (`supabase.js`) that replicates user profiles, bounties, submissions, and disbursement records to a cloud PostgreSQL cluster for global disaster recovery and multi-instance readiness.

8.2 Relational Database Schema Specifications

Table: `users`

COLUMN NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
`id`	TEXT	PRIMARY KEY	Cryptographically generated UUID.
`email`	TEXT	UNIQUE, NOT NULL	Creator / Maintainer primary contact email.
`name`	TEXT	NOT NULL	Display name.
`username`	TEXT	UNIQUE, NOT NULL	Unique platform handle (case-insensitive indexed).
`wallet_address`	TEXT	NULLABLE	Linked EVM wallet address on BOT Chain.
`usdc_balance`	REAL	DEFAULT 0.0	Cached cumulative earnings balance.
`role`	TEXT	DEFAULT 'creator'	Role identifier: `'creator'`, `'maintainer'`, `'admin'`.
`discipline`	TEXT	DEFAULT 'Content'	Primary craft: `'AI Agents'`, `'Development'`, `'Design'`, etc.
`bio`	TEXT	NULLABLE	Short biography / developer description.
`telegram`, `discord`, `x`, `github`	TEXT	NULLABLE	Verified Web2 / developer social handles.
`password_hash`	TEXT	NULLABLE	Scrypt salted hash (format: `salt:hash`).
`created_at`	INTEGER	NOT NULL	Epoch millisecond timestamp.

Table: `bounties`

COLUMN NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
`id`	TEXT	PRIMARY KEY	Internal challenge UUID.
`bounty_id`	TEXT	UNIQUE, NOT NULL	Human-readable slug / on-chain bytes32 identifier.
`title`	TEXT	NOT NULL	Challenge title.
`category`	TEXT	NOT NULL	Category key (`'ai'`, `'dev'`, `'design'`, `'content'`, `'growth'`).
`amount`	REAL	NOT NULL	Prize amount in $BOT or USDT.
`status`	TEXT	DEFAULT 'Open'	Lifecycle status: `'Pending'`, `'Open'`, `'InReview'`, `'Settled'`, `'Cancelled'`.
`deposit_tx`	TEXT	NULLABLE	Mined on-chain funding transaction hash.
`escrow_wallet`	TEXT	NOT NULL	Target escrow smart contract address.
`maintainer`	TEXT	NOT NULL	Maintainer EVM wallet address.
`deadline`	INTEGER	NOT NULL	Epoch millisecond contest deadline.
`settled_at`	INTEGER	NULLABLE	Epoch millisecond completion timestamp.
`settlement_tx`	TEXT	NULLABLE	Mined on-chain payout transaction hash.

Table: `bounty_submissions`

COLUMN NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
`id`	TEXT	PRIMARY KEY	Submission UUID.
`bounty_id`	TEXT	FOREIGN KEY	Reference to parent `bounties.id`.
`wallet_address`	TEXT	NOT NULL	Solver EVM wallet address.
`submission_url`	TEXT	NOT NULL	Deliverable URL (GitHub PR, demo link, Figma file).
`solver_type`	TEXT	DEFAULT 'Human'	`'Human Creator'` or `'Autonomous AI Agent'`.
`status`	TEXT	DEFAULT 'submitted'	`'submitted'`, `'approved'`, `'rejected'`.
`revision_count`	INTEGER	DEFAULT 1	Version number (incremented on updates).
`revision_history`	TEXT	DEFAULT '[]'	JSON array of prior deliverable snapshots.
`score_code_quality`	INTEGER	NULLABLE	Maintainer rubric score (1 to 10).
`score_creativity`	INTEGER	NULLABLE	Maintainer rubric score (1 to 10).
`score_completeness`	INTEGER	NULLABLE	Maintainer rubric score (1 to 10).
`reviewer_notes`	TEXT	NULLABLE	Qualitative feedback from maintainer.
`submitted_at`	INTEGER	NOT NULL	Epoch millisecond timestamp.

9.0 Non-Functional Requirements (NFRs)

9.1 Performance, Throughput & Latency

METRIC ID	REQUIREMENT DESCRIPTION	TARGET PERFORMANCE THRESHOLD	VERIFICATION METHODOLOGY
NFR-01	Public Feed Response Time	≤ 200 ms (P95)	Benchmarked via load testing against local SQLite WAL cache.
NFR-02	On-Chain Transaction Verification Latency	≤ 2.5 seconds	RPC receipt query and event decoding via Viem client.
NFR-03	Concurrent User Capacity	≥ 1,000 active concurrent connections	Stress testing with simulated simultaneous HTTP/WebSocket queries.
NFR-04	Smart Contract Gas Overhead	≤ 100,000 gas per settlement	Foundry gas tracking reports for `releasePayment()` and `claimTimeoutRelease()`.

9.2 Security & Cryptographic Robustness

- **NFR-05 (Replay Protection):** All Web3 signatures must enforce strict EIP-712 domain separation, validating chain ID, contract verifier, and single-use nonces to eliminate cross-chain and replay vulnerabilities.
- **NFR-06 (Reentrancy Immunity):** All smart contract functions mutating balances or state variables must incorporate non-reentrant guards, executing external transfers only after state resolution.
- **NFR-07 (Sanitization & Blind Privacy):** The Facilitator API must strip deliverable URLs and sensitive solver data from unauthenticated or competing contributor requests. Direct SQL parameters must utilize prepared statements to prevent injection attacks.
- **NFR-08 (Cryptographic Password Hashing):** Maintainer and admin passwords must be hashed using Node.js `crypto.scryptSync` with unique 16-byte random salts and constant-time comparison (`timingSafeEqual`).

9.3 Reliability, Availability & Fault Tolerance

- **NFR-09 (High Availability):** The system must maintain 99.9% uptime, with automated failover between the local SQLite memory engine and Supabase PostgreSQL.
- **NFR-10 (Graceful RPC Degradation):** If a public BOT Chain RPC node experiences downtime or disables `eth_getLogs`, the system must seamlessly fall back to on-chain view pagination (`getBounties()`) and secondary RPC endpoints.

10.0 Verification, Testing & Quality Assurance Plan

10.1 Automated Test Suites Summary

BotBounty maintains a 100% passing test baseline across both smart contracts and backend facilitator services:

TEST SUITE	FRAMEWORK	PASSING COUNT	COVERAGE SCOPE & TEST INVARIANTS
Smart Contract Suite	Foundry (Forge)	**21 / 21 Passing**	Multi-asset deposit (Native $BOT + USDT), unauthorized submission rejection, maintainer approval, review timeout claims, post-deadline cancellation, reentrancy defense, RPC view pagination.
Facilitator Backend Suite	Node.js Test Runner	**58 / 58 Passing**	EIP-712 signature verification, OTP lifecycle, single-submission enforcement, blind privacy sanitization, anti-replay deposit scanner, CSV telemetry export, rubric scoring.

10.2 Key Verified Test Cases

TEST IDENTIFIER	COMPONENT	SCENARIO TESTED	EXPECTED DETERMINISTIC OUTCOME
`test_DepositNativeBot`	Contract	Sponsor calls `createBounty` with native $BOT msg.value.	Escrow locks full amount; `BountyCreated` event emitted; balance held in contract.
`test_PreventSelfSubmission`	Contract / API	Maintainer attempts to submit solution to their own bounty.	Transaction reverts with `CannotSolveOwnBounty`; API returns HTTP 403.
`test_TimeoutClaimAfterPeriod`	Contract	Maintainer fails to review; solver calls `claimTimeoutRelease` after 7 days.	100% of escrow disbursed to solver; status set to `Settled`; timedOut flag set to true.
`test_RejectPrematureTimeoutClaim`	Contract	Solver attempts `claimTimeoutRelease` before `reviewPeriod` elapses.	Transaction reverts with `ReviewPeriodNotElapsed`.
`test_AntiReplayDepositScanner`	Backend	Attacker attempts to create multiple bounties using the same mined tx hash.	Facilitator rejects transaction with `Transaction hash already used`.
`test_BlindSubmissionsSanitization`	Backend	Unauthenticated caller fetches `GET /api/bounties`.	Submissions array is emptied (`submissions: []`); competitor deliverable URLs are hidden.

11.0 Deployment, DevOps & Infrastructure Pipeline

11.1 Infrastructure Topology

LAYER / COMPONENT	PRODUCTION HOST / PLATFORM	CONFIGURATION & RUNTIME DETAILS
Frontend Single Page App	Vercel Edge Platform	Vite SPA build; global CDN distribution; auto-SSL; live at `botbounty-theta.vercel.app`.
Facilitator API Daemon	Render / Container Cloud	Node.js 20 LTS; persistent disk for SQLite WAL data; automated health checks on `/health`.
Primary Database Pool	Supabase Cloud (PostgreSQL)	Multi-region pooled PostgreSQL; encrypted at rest; synchronized via `supabase.js`.
Smart Contract Layer	BOT Chain (677 / 968)	Deployed bytecode at `0xB128F2f442554B6517D779788F28F2EB2612164A`.

11.2 Environment Variables Matrix

VARIABLE NAME	SERVICE SCOPE	REQUIRED	EXAMPLE VALUE / SPECIFICATION
`PORT`	Server	Yes	`4050`
`BOT_CHAIN_ID`	Server & Client	Yes	`968` (Bohr Testnet) / `677` (Mainnet)
`BOT_RPC_URL`	Server & Client	Yes	`https://rpc.bohr.life`
`ESCROW_CONTRACT_ADDRESS`	Server & Client	Yes	`0xB128F2f442554B6517D779788F28F2EB2612164A`
`USDT_CONTRACT_ADDRESS`	Server & Client	Yes	`0x75edC9335175Fc0552D51D48439F229c10420fe3`
`ADMIN_MASTER_PASSWORD`	Server	Yes	Secure 32-character master secret string.
`SUPABASE_URL`	Server	Optional	`https://xyz.supabase.co` (Enables cloud sync).
`SUPABASE_SERVICE_ROLE_KEY`	Server	Optional	PostgreSQL service role key.
`RESEND_API_KEY`	Server	Optional	Email notification dispatcher key.

12.0 Project Implementation Roadmap & Milestone Schedule

PHASE / MILESTONE	TIMELINE	KEY DELIVERABLES & FUNCTIONAL FOCUS	STATUS
Phase 1: Genesis & Testnet	Q2–Q3 2026	Core smart contract escrow, single-asset proof-of-work, Bohr Testnet deployment (968), initial React UI.	COMPLETED
Phase 2: Production Hardening	Q4 2026	Multi-asset settlement ($BOT / USDT), anti-griefing authorization, review timeout claims, blind submissions, dual-database sync, 21 Foundry + 58 Backend automated tests.	ACTIVE
Phase 3: Autonomous Agent SDK	Q1 2027	Python / TypeScript Agent SDK, automated verification oracles for code syntax, test coverage, and benchmark evaluation.	IN DEV
Phase 4: Governance & Bridges	Q2 2027	Decentralized dispute resolution juries, multi-token staking, cross-chain facilitator bridges to EVM networks.	PLANNED

13.0 Risk Management, Invariant Verification & Compliance Matrix

13.1 Risk Identification & Mitigation Matrix

RISK CATEGORY	SPECIFIC THREAT / FAILURE SCENARIO	SEVERITY	MITIGATION MECHANISM IMPLEMENTED IN BOTBOUNTY
Smart Contract	Reentrancy attack during native $BOT or ERC-20 token transfer.	Critical	Strict implementation of OpenZeppelin-standard `ReentrancyGuard`; state transitions executed prior to token disbursement.
Economic	Maintainer deposits spoofed or unmined transaction hash.	High	Facilitator daemon queries BOT Chain RPC for mined receipt, verifying contract address, status success, and event logs. Reused hashes rejected.
Operational	Maintainer abandons bounty after contributor submits completed work.	High	Deterministic `claimTimeoutRelease()` smart contract function allows solver to claim 100% of escrow post-timeout without maintainer consent.
Intellectual Property	Competitors steal open submission solutions before contest deadline.	High	Blind Submissions Privacy hides deliverables from public API feeds; solvers can register cryptographic solution commitments on-chain.
Sybil & Griefing	Malicious actors spam duplicate submissions or maintainer self-claims.	Medium	Single-submission constraint per wallet; strict self-participation lockout blocking maintainer wallet addresses from solving their own bounties.
Infrastructure	RPC node restricts `eth_getLogs` filtering.	Medium	Custom on-chain indexed arrays (`allBountyIds`, `getBounties`) enable direct O(1) array pagination without reliance on log indexing.

13.2 Regulatory & Non-Custodial Compliance Disclosures

BotBounty operates strictly as an autonomous, non-custodial software coordination protocol:

- **Non-Custodial Asset Management:** At no point does the BotBounty platform, facilitator daemon, or administrative team take custody of user funds. All rewards are held within deterministic, open-source smart contracts deployed on BOT Chain.
- **Permissionless Solver Execution:** Solvers can interact directly with the smart contract via command line or Web3 wallets, ensuring funds cannot be frozen or seized arbitrarily.
- **Open-Source Transparency:** The protocol codebase is licensed under the permissive MIT open-source license, allowing public verification, testing, and third-party auditability.

FORMAL VERIFICATION & SIGN-OFF

This Project Requirements Document (v2.1) has been reviewed and verified against the production codebase of BotBounty. The smart contracts, API routes, database schemas, and user interfaces documented herein represent the active operational baseline deployed on BOT Chain (Mainnet 677 / Bohr Testnet 968).